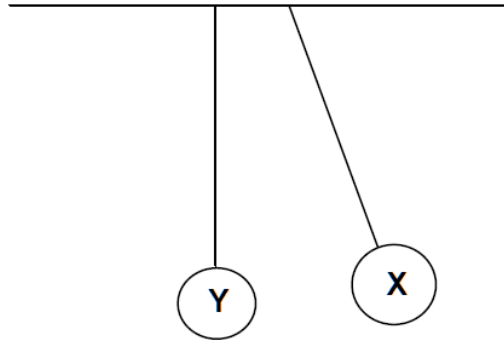


- 5 Two steel balls X and Y are suspended on strings. Ball X is pulled to one side as shown.



After ball X is released, the balls collide.

Which quantities must be conserved in the collision?

- A kinetic energy, total energy and momentum
- B kinetic energy and momentum only
- C kinetic energy and total energy only
- D total energy and momentum only

Ans: D

Collisions could be categorised as elastic or inelastic. Only in elastic collisions are kinetic energy conserved. Hence, only total energy (i.e. including sound, light, heat, etc.) and momentum, must be conserved.

- 6 In reverse bungee jumping, passengers in a spherical cage are pulled upwards by